

Full solution packet for arXiv:2205.06839

Arbitrary-weight disjoint superdemocracy implies superdemocracy

Status. full_solution_likely_valid. This packet answers Question 2 in Section 6 of Hùng Việt Chu, *On weighted greedy-type bases*, arXiv:2205.06839 [1].

Source question. In Definition 1.2, a set weight is an arbitrary map $w : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty]$ such that $w(\emptyset) = 0$ and $w(A) > 0$ for every nonempty A . Question 2 on source page 14 asks:

For weights in Definition 1.2, is an ω -disjoint superdemocratic basis also ω -superdemocratic? If not, what minimal condition(s) to put on ω so that the two properties are equivalent?

ω -partially greedy, it is quasi-greedy by Theorem 5.3 and has ς -left-Property (A) for $s(n) = 2^{-n}$. Therefore, the basis is automatically ς -partially greedy.

6. QUESTIONS AND DISCUSSION

We list several open questions for future research.

- Q1 We show that for a structured weight ω , a basis is ω -almost greedy if and only if it is ω -semi-greedy. Does the result hold for a larger class of weights?
- Q2 For weights in Definition 1.2, is an ω -disjoint superdemocratic basis also ω -superdemocratic? If not, what minimal condition(s) to put on ω so that the two properties are equivalent.

For the second question, we know that for a structured weight, an ω -disjoint superdemocratic is ω -superdemocratic.

Definitions. Let $\mathcal{B} = (e_n)_{n=1}^\infty$ be the semi-normalized Schauder basis fixed in the source paper, with biorthogonal functionals (e_n^*) . Put

$$M = \sup_n \|e_n\| < \infty, \quad M^* = \sup_n \|e_n^*\| < \infty.$$

For a finite set A and signs $\varepsilon = (\varepsilon_n)_{n \in A}$, where $|\varepsilon_n| = 1$, write

$$\mathbf{1}_{\varepsilon A} = \sum_{n \in A} \varepsilon_n e_n.$$

The basis is K - ω -disjoint superdemocratic if

$$\|\mathbf{1}_{\varepsilon A}\| \leq K \|\mathbf{1}_{\delta B}\| \tag{1}$$

whenever $A, B \subset \mathbb{N}$ are finite, $A \cap B = \emptyset$, $w(A) \leq w(B)$, and the signs are arbitrary. It is ω -superdemocratic when the same comparison holds without the disjointness requirement.

Proof intuition. The weight itself need not supply a fresh set of intermediate weight, so the source proof for structured weights cannot be copied. Instead, we make a fresh set of intermediate norm. If signed constant-coefficient sums are unbounded, then they remain unbounded after deleting any finite set of coordinates. Adding their terms one at a time gives a signed sum whose norm lies just above any prescribed level. For two putative counterexample sums of norms $a \gg b$, choose a fresh sum of norm strictly between Kb and a/K . The disjoint comparison with the first set forces the fresh set's weight below the first weight. Transitivity places it below the second weight, and

the disjoint comparison with the second set then contradicts its norm being greater than Kb . If all signed sums are bounded instead, the conclusion follows immediately from the uniform lower bound on every nonzero signed sum.

Bridge lemma. *Suppose*

$$\sup\{\|\mathbf{1}_{\varepsilon E}\| : E \subset \mathbb{N} \text{ finite, } |\varepsilon_n| = 1\} = \infty. \quad (2)$$

For every finite $F \subset \mathbb{N}$ and every $t \geq 0$, there are a nonempty finite $C \subset \mathbb{N} \setminus F$ and signs η such that

$$t < \|\mathbf{1}_{\eta C}\| \leq t + M. \quad (3)$$

Proof. First, signed sums supported in $\mathbb{N} \setminus F$ are unbounded. Indeed, if they were bounded by L , then every signed sum would satisfy

$$\|\mathbf{1}_{\varepsilon E}\| \leq \sum_{n \in E \cap F} \|e_n\| + \|\mathbf{1}_{\varepsilon(E \setminus F)}\| \leq |F|M + L,$$

contrary to (2).

Choose a finite $D \subset \mathbb{N} \setminus F$ and signs η such that $\|\mathbf{1}_{\eta D}\| > t + M$. Enumerate $D = \{n_1, \dots, n_m\}$ and let $z_j = \sum_{i=1}^j \eta_{n_i} e_{n_i}$, with $z_0 = 0$. Let j be the first index for which $\|z_j\| > t$. Then $\|z_{j-1}\| \leq t$, and hence

$$t < \|z_j\| \leq \|z_{j-1}\| + \|e_{n_j}\| \leq t + M.$$

Taking $C = \{n_1, \dots, n_j\}$ proves the claim. $\square$

Theorem (Affirmative answer to Question 2). *Let $w : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty]$ be any weight in Definition 1.2 of the source paper. Every ω -disjoint superdemocratic Schauder basis is ω -superdemocratic. No additional condition on w is required.*

More precisely, suppose (1) holds with constant K , and write

$$D = \sup\{\|\mathbf{1}_{\varepsilon E}\| : E \subset \mathbb{N} \text{ finite, } |\varepsilon_n| = 1\}.$$

If $D < \infty$, a superdemocracy constant is DM^ . If $D = \infty$, a superdemocracy constant is*

$$K^2 + KMM^*. \quad (4)$$

Proof. Let $A, B \subset \mathbb{N}$ be finite, let $w(A) \leq w(B)$, and fix signs ε, δ . We must bound

$$a = \|\mathbf{1}_{\varepsilon A}\| \quad \text{by a constant times} \quad b = \|\mathbf{1}_{\delta B}\|.$$

If $A = \emptyset$, there is nothing to prove. If $A \neq \emptyset$, then $B \neq \emptyset$, because $w(A) > 0 = w(\emptyset)$. For any $j \in B$, biorthogonality gives

$$1 = |e_j^*(\mathbf{1}_{\delta B})| \leq \|e_j^*\|b \leq M^*b, \quad \text{so} \quad b \geq (M^*)^{-1}. \quad (5)$$

If $D < \infty$, then $a \leq D \leq DM^*b$, proving the claim in this case. Notice that this argument does not use the weight at all.

Now assume $D = \infty$. We claim that

$$a \leq (K^2 + KMM^*)b.$$

Suppose otherwise. By (5),

$$a > (K^2 + KMM^*)b \geq K^2b + KM = K(Kb + M). \quad (6)$$

Apply the bridge lemma to $F = A \cup B$ and $t = Kb$. We obtain a finite $C \subset \mathbb{N} \setminus (A \cup B)$ and signs η such that, on writing $c = \|\mathbf{1}_{\eta C}\|$,

$$Kb < c \leq Kb + M < a/K, \quad (7)$$

where the last inequality is (6).

If $w(A) \leq w(C)$, then $A \cap C = \emptyset$ and (1) would give $a \leq Kc < a$, a contradiction. Since the extended real numbers are totally ordered, we must therefore have

$$w(C) < w(A) \leq w(B).$$

But $C \cap B = \emptyset$, so another application of (1) gives $c \leq Kb$, contradicting (7). This proves (4) and completes the proof. $\square$

Consequences. The answer to the first sentence of Question 2 is yes. Therefore the second sentence has the strongest possible answer: within Definition 1.2, there are no additional conditions to impose on ω . Proposition 6.1 of the source paper, which proves the implication for structured weights, follows as a special case.

The proof in fact needs less than the source’s Schauder-basis hypotheses. It uses neither convergence of basis expansions nor boundedness of partial-sum projections. Consequently, the same result holds for every biorthogonal system (e_n, e_n^*) for which both $\sup_n \|e_n\|$ and $\sup_n \|e_n^*\|$ are finite.

Verification notes. The proof uses only four structural facts: finite supports have an infinite complement; $\sup_n \|e_n\| < \infty$; $\sup_n \|e_n^*\| < \infty$; and the total order on $[0, \infty]$. Thus weights equal to ∞ cause no exception. The bounded case and unbounded case exhaust all possibilities. The bridge construction is valid over both $\mathbb{R}$ and $\mathbb{C}$, since multiplying a basis vector by a scalar of modulus one does not alter its norm.

The run’s lightweight indexes and local arXiv source corpus were searched for arXiv:2205.06839, the title, the exact Question 2 wording, and close variants of “disjoint superdemocratic” with arbitrary set weights. Bounded web and forward-citation searches found no exact later solution. Crossref and OpenAlex returned no citing works for the published DOI; Semantic Scholar’s sole result was checked directly and contains only a commented-out reference, with no discussion of Question 2. This is a bounded rather than exhaustive search, so novelty confidence is moderate-to-high pending expert review; full details are in `citation_audit.md`.

Limitations. This packet answers Question 2 only. It does not address Question 1 about extending the equivalence of ω -almost greedy and ω -semi-greedy bases beyond structured weights. The quantitative constant in the bounded-sum case depends on D ; no optimality claim is made for either displayed constant.

Human-review recommendation. Review as a candidate full solution. The most important checks are (i) the claim that unbounded signed sums remain unbounded after deleting finitely many coordinates, and (ii) the two weight-order deductions following (7). Both are explicit in the proof.

References

- [1] H. V. Chu, *On weighted greedy-type bases*, Bulletin of the Brazilian Mathematical Society, New Series **54** (2023), Article 49; arXiv:2205.06839; DOI:10.1007/s00574-023-00367-3.